TAN TAO UNIVERSITY

SCHOOL OF INFORMATION TECHNOLOGY

Midterm Project - Predict Customer Churn

Course: Data Visualization (CS441V)

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Group 1

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Github: MinhThuan/Data Visualization Course (CS441V)

1 Introduce

1.1 Define a Problem

Customer churn is a major challenge for companies that provide subscription-based services. When customers stop using a service, companies may lose revenue and incur additional costs to acquire new customers. Therefore, understanding the factors associated with customer churn is important for improving customer retention strategies.

The objective of this project is to explore which customer characteristics are associated with a higher likelihood of churn. The dataset contains multiple features describing customer demographics, service usage, and billing information, such as contract type, tenure, monthly charges, and internet services.

Using data visualization techniques, this project analyzes how different features are distributed between customers who churn and those who remain with the service. By examining these patterns, we aim to identify which characteristics may indicate a higher risk of customer churn.

- Which customer characteristics are associated with a higher likelihood of churn?
- How do features such as contract type, tenure, and monthly charges influence the probability of churn?

Through visual exploration of the dataset, the project aims to better understand the factors that influence customer churn and identify patterns that may help detect customers at risk of leaving the service.

2 Dataset Description

2.1 Dataset Source

The dataset used in this project is the *Customer Churn Dataset*, which is publicly available on Kaggle. This dataset contains information about customers of a telecommunications company and whether they have discontinued the service.

The dataset includes various attributes related to customer demographics, account information, and service usage. These features help analyze the characteristics of customers who are more likely to churn.

2.2 Dataset Overview

The dataset contains multiple records representing individual customers. Each record describes a customer's profile, including demographic information, services subscribed to, billing details, and the churn status.

The target variable in the dataset is **Churn**, which indicates whether a customer has left the service.

- **Churn = 0**: Customer with low probability of churn (customer likely to stay)
- **Churn = 1**: Customer with high probability of churn (customer likely to leave)

2.3 Feature Description

Table 2.3 summarizes several important features used in the analysis.

Feature	**Description**
gender	Gender of the customer
SeniorCitizen	Whether the customer is a senior citizen
Partner	Whether the customer has a partner
Dependents	Whether the customer has dependents
tenure	Number of months the customer has stayed with the company
PhoneService	Whether the customer has phone service
InternetService	Type of internet service
Contract	Type of contract (Month-to-month, One year, Two year)
MonthlyCharges	The amount charged to the customer monthly
TotalCharges	The total amount charged to the customer
Churn	Whether the customer left the service

2.4 Data Preparation

Before performing data visualization, several preprocessing steps were applied to the dataset. These steps include removing unnecessary columns, converting data types for numerical variables, and encoding the churn variable into binary values. These preprocessing steps ensure that the dataset is suitable for analysis and visualization.

3 Data Visualization and Analysis

To better understand the factors associated with customer churn, several data visualization techniques were applied to explore the distribution of key numerical features. In particular, the analysis focuses on three important variables: *tenure*, *MonthlyCharges*, and *TotalCharges*. Kernel Density Estimation (KDE) plots are used to compare the probability distributions of these features for customers who stayed and customers who churned.

3.1 Tenure and Customer Churn

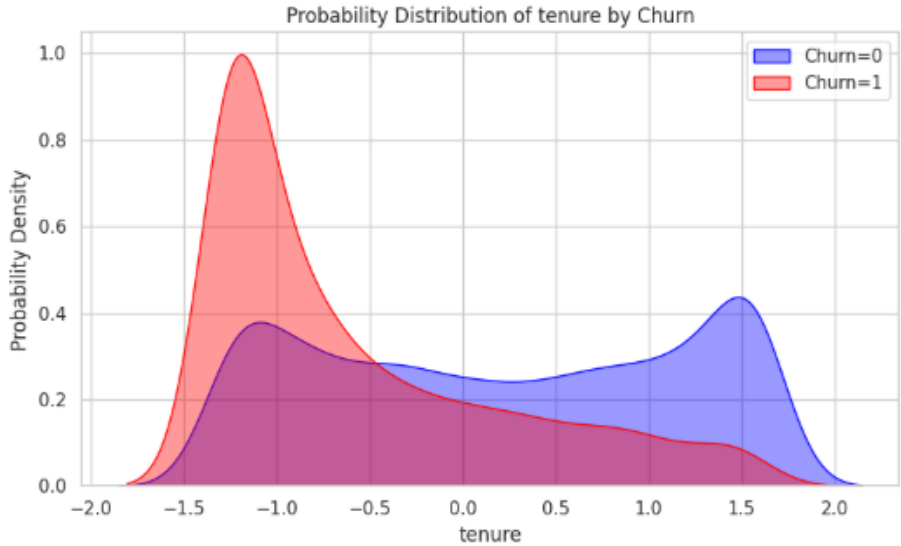

Figure 1: Probability distribution of tenure by churn

Figure 1 illustrates the probability distribution of the *tenure* variable for customers who stayed and those who churned.

The visualization shows a clear difference between the two groups. Customers who churned tend to have significantly shorter tenure values compared to those who remained with the service. The density for churned customers is concentrated toward lower tenure values, indicating that many customers leave the service within the early months.

In contrast, customers who stayed with the company generally have higher tenure values, suggesting that long-term customers are less likely to churn.

3.2 Monthly Charges and Customer Churn

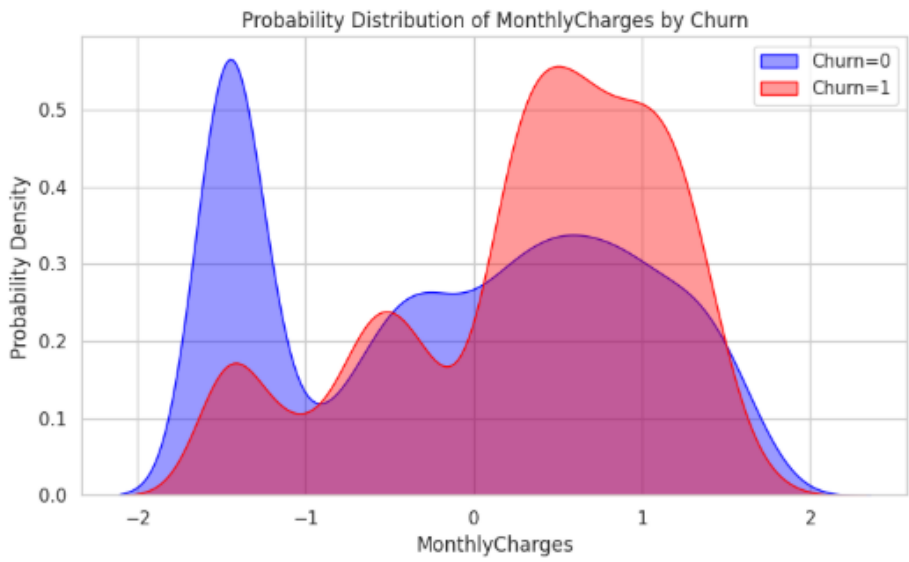

Figure 2: Probability distribution of MonthlyCharges by churn

Figure 2 presents the probability distribution of *MonthlyCharges* for both churned and non-churned customers.

Based on the visualization, customers who pay lower monthly charges tend to remain with the service for a longer period (Churn = 0). The density for non-churned customers is higher in the lower monthly charge range, indicating that customers using basic or lower-cost service plans are more likely to continue using the service.

On the other hand, customers with higher monthly charges show a greater tendency to churn. The density for churned customers is concentrated in the higher monthly charge range. This suggests that customers who subscribe to higher-priced or premium services may be more likely to discontinue the service. One possible explanation is that these higher-cost services may not fully meet customer expectations, leading them to stop using the service.

Overall, the visualization suggests that lower-cost services tend to maintain more stable and long-term customers, while higher-cost services may have a higher risk of customer churn.

3.3 Total Charges and Customer Churn

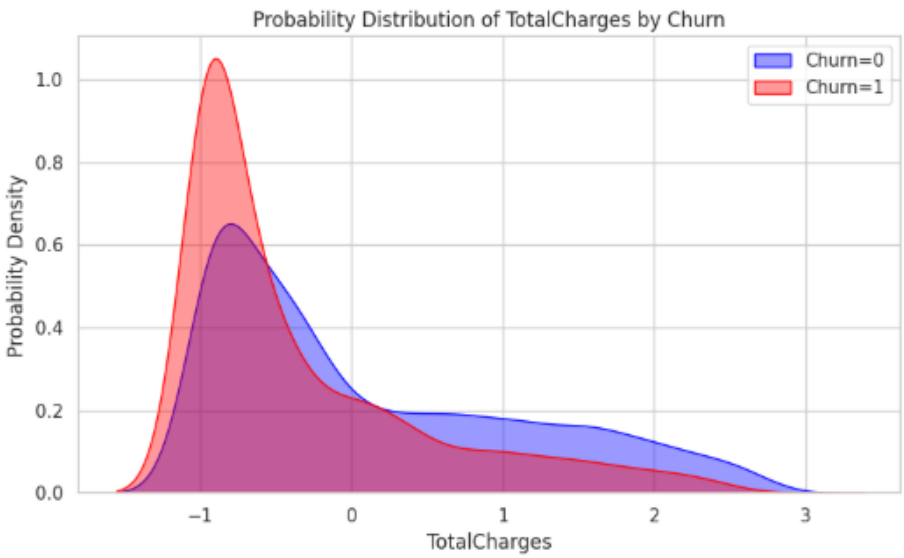

Figure 3: Probability distribution of TotalCharges by churn

Figure 3 shows the probability distribution of *TotalCharges* for both groups of customers.

From the visualization, customers who churned (Churn = 1) have a high density in the lower range of total charges. This indicates that many customers who leave the service tend to do so after paying a relatively small total amount, suggesting that they stop using the service early.

As the total charges increase, the density of churned customers gradually decreases, meaning fewer customers continue using the service long enough to accumulate higher total payments before leaving.

In contrast, customers who did not churn (Churn = 0) initially show lower density in the lower total charge range. However, as the total charges increase, the density of non-churned customers becomes higher compared to churned customers. This suggests that customers who continue using the service tend to accumulate higher total payments over time.

Overall, the visualization indicates that services with moderate to lower total costs tend to maintain customers more sustainably, and more customers continue using services within this range over time.

3.4 Summary of Findings

From the visual analysis of the three key features (*tenure*, *MonthlyCharges*, and *TotalCharges*), several important patterns can be observed.

- Customers with shorter tenure are more likely to churn, indicating that many customers leave the service during the early stages of usage.
- Customers who pay lower monthly charges tend to remain with the service for a longer period. In contrast, customers who subscribe to higher-priced or premium services show a greater tendency to churn.

- Customers who churn usually accumulate lower total charges because they discontinue the service earlier. Meanwhile, customers who stay with the service tend to accumulate higher total payments over time.

Overall, the analysis suggests that lower or moderate-cost services tend to maintain more stable and long-term customers, while higher-priced services may have a higher risk of customer churn. These findings highlight the importance of improving customer experience and service value, especially for higher-cost service plans, in order to reduce churn and maintain long-term customer relationships.

4 Conclusion

This project focused on analyzing customer churn and identifying key factors that influence whether a customer is likely to leave the service. Using the customer churn dataset, several preprocessing steps were performed, including data cleaning, handling missing values, encoding categorical variables, and standardizing numerical features.

Through data visualization and exploratory analysis, important patterns related to customer churn were identified. In particular, three numerical variables — *tenure*, *MonthlyCharges*, and *TotalCharges* — were found to have significant relationships with churn behavior.

The analysis revealed that customers with shorter tenure are more likely to leave the service, suggesting that the early stage of customer engagement is a critical period for retention. Additionally, customers with higher monthly charges show a higher tendency to churn, indicating that pricing may influence customer decisions. Finally, customers who churn generally have lower total charges because they discontinue the service earlier.

These findings highlight the importance of customer retention strategies, especially during the early months of service usage. Companies can reduce churn by improving onboarding experiences, offering promotional pricing, or providing incentives for long-term subscriptions.

Overall, this study demonstrates how data preprocessing, visualization, and exploratory analysis can provide valuable insights into customer behavior and support decision-making in customer relationship management.

5 Appendix: Source Code

```python
# -*- coding: utf-8 -*-
"""Midterm_Project_Predict_Customer_Churn.ipynb

Automatically generated by Colab.

Original file is located at
    https://colab.research.google.com/drive/1EDBs-fJ-HOZspb_3V3Uqyd2He-zQLaud
"""

import kagglehub

# Download dataset
path = kagglehub.dataset_download(
    "mohankrishnathalla/predict-customer-churn-submission-dataset"
)

print("Dataset path:", path)

import os

files = os.listdir(path)
print(files)

import pandas as pd

# Read the Origin.csv file from the downloaded path
df = pd.read_csv(f"{path}/Original.csv")

# Check the dataset shape (number of rows and columns)
print("Shape:", df.shape)

# Display the first few rows of the dataset
print(df.head())

# Convert the dataset into feature vectors

import pandas as pd
from sklearn.preprocessing import StandardScaler

# Remove unnecessary column
df = df.drop(columns=["customerID"])

# Convert TotalCharges to numeric and handle missing values
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df["TotalCharges"] = df["TotalCharges"].fillna(df["TotalCharges"].median())
```

```python

# Encode the target variable
df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})

# Encode binary Yes/No variables (except gender)
binary_cols = ["Partner","Dependents","PhoneService",
               "OnlineSecurity","OnlineBackup","DeviceProtection",
               "TechSupport","StreamingTV","StreamingMovies","PaperlessBilling"]

for col in binary_cols:
    df[col] = df[col].map({"Yes":1,"No":0})

# Handle gender separately (Male/Female)
df["gender"] = df["gender"].map({"Female":0, "Male":1})

# Apply One-Hot Encoding for categorical variables with multiple categories
multi_cat_cols = ["MultipleLines","InternetService","Contract","PaymentMethod"]
df = pd.get_dummies(df, columns=multi_cat_cols, drop_first=True)

# Normalize numerical features
numeric_cols = ["SeniorCitizen","tenure","MonthlyCharges","TotalCharges"]
scaler = StandardScaler()
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])

# Split into feature vectors X and target variable y
X = df.drop(columns=["Churn"])
y = df["Churn"]

print("Shape X:", X.shape)
print("Shape y:", y.shape)
X.head()

import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

# Split the dataset based on churn value
df_churn0 = df[df["Churn"] == 0]
df_churn1 = df[df["Churn"] == 1]

# Get the list of feature columns (excluding the target column)
feature_cols = [col for col in df.columns if col != "Churn"]

```

```python
for col in feature_cols:
    plt.figure(figsize=(8,5))

    # Check if the feature is categorical (few unique values)
    if df[col].nunique() <= 10:
        # Calculate the percentage proportion for each churn group
        prop0 = df_churn0[col].value_counts(normalize=True).reset_index()
        prop0.columns = [col, "proportion"]
        prop0["Churn"] = 0

        prop1 = df_churn1[col].value_counts(normalize=True).reset_index()
        prop1.columns = [col, "proportion"]
        prop1["Churn"] = 1

        # Combine the two groups into one dataframe
        prop_df = pd.concat([prop0, prop1])

        # Plot bar chart for categorical variables
        sns.barplot(x=col, y="proportion", hue="Churn", data=prop_df, palette="Set2")
        plt.ylabel("Rate (%)")

    else:
        # Plot KDE distribution for continuous variables
        sns.kdeplot(data=df_churn0, x=col, fill=True, label="Churn=0", alpha=0.4, color="blue")
        sns.kdeplot(data=df_churn1, x=col, fill=True, label="Churn=1", alpha=0.4, color="red")
        plt.ylabel("Probability Density")
        plt.legend()

    # Add chart title
    plt.title(f"Probability Distribution of {col} by Churn")
    plt.tight_layout()
    plt.show()
```

Author Contributions

This project was completed by two team members with the following responsibilities:

Trieu Minh Thuan

- Data preprocessing and data cleaning
- Supporting data analysis
- Writing the report (Introduction, Data Analysis, Conclusion)

Ho Nguyen Kim Long

- Problem formulation and dataset description
- Handling missing values and feature transformation
- Interpretation of visualization results
- Implementing visualization code